工程數學

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⑧(Exponential forcing)

$$y' + y = e^{2x}$$

⑩ mixed forcing

$$y'' - y = e^{x} + x$$

大綱

1. First Order Derivative(8)
2. Second Order Derivative(10)
3. 可分離微分方程 separable equation (optional)

First Order Derivative(8)

First Order Derivative(8)

①

$$y' + y = e^{2x}$$

②

Pattern : $y' + P(x)y = Q(x)$

名: first-order ordinary differential equation.

一階常微分方程。

③

$$I(x) = e^{\int P(x)dx} \Rightarrow 代入 \; P(x) = 1$$

$$I(x) = e^{\int 1 dx} = e^{x}$$

④

將 e^{x} 代入: $e^{x}y' + e^{x}y = e^{x}e^{2x}$

$$e^{x}y' + e^{x}y = e^{2x+x} = e^{3x}$$

⑤ 積分:

$$\int \frac{d}{dx}(e^{x}y)\,dx = \int e^{3x}dx$$

$$\frac{1}{1}e^{x}y = \frac{1}{3}e^{3x} + C$$

First Order Derivative(8)

① 分離變數：同除 e^x

$$e^x y = \frac{1}{3}e^{3x} + C \Rightarrow y = \frac{1}{3}e^{3x} \times e^{-x} + C \times e^{-x}$$

Answer: $y = \frac{1}{3}e^{2x} + Ce^{-x}$

$\Rightarrow$ $\div e^x \Rightarrow$ 同 $\Rightarrow \times e^{-x}$

而: $e^{3x} \times e^{-x} = e^{3x-x} = e^{2x}$

Second Order Derivative(10)

Second Order Derivative(10)

① $y'' - y = e^x + x$

② $y'' - y = 0$

則: $r_1 = 1$, $r_2 = -1$, 因 y'' 正 $(-y)$ 負

③ $r^2 - 1 = 0$ & $(r-1)(r+1) = 0$

$1^2 - 1 = 0$, $(1-1)(1+1) = 0 \times 2 = 0$

$(-1)^2 - 1 = 0$, $(-1-1)(-1+1) = -2 \times 0 = 0$

均質

Second Order Derivative(10)

4 $y_h = C_1 e^{x} + C_2 e^{-x}$

$\boxed{因}$ $r_1 = 1,\ r_2 = -1$

5 非均質部分:

$g(x) = e^{x} + x$ ([illegible])

6 拆分: $g(x) = e^{x} + x$

$g_1(x) = e^{x}$, $g_2(x) = x$

7 $g_1(x) = e^{x} \Rightarrow Axe^{x} = y_{p1}$

$g_2(x) = x \Rightarrow Bx + D = y_{p2}$

$\Rightarrow y_{p1} + y_{p2} = Axe^{x} + Bx + D = y_p$

特解

8 $y_p = Axe^{x} + Bx + D$

$y_p'' = A(e^{x} + e^{x} + xe^{x}) = A(2e^{x} + xe^{x})$

$y_p' = A(e^{x} + xe^{x}) + B = \cancel{A}$ …

Second Order Derivative(10)

(9) 替换: $y'' - y \Rightarrow (2Ae^x + Axe^x)$

$- (Axe^x + Bx + D) = e^x + x$

&

$2Ae^x + (-Bx) - D = e^x + x$

(10) $2A = 1 \Rightarrow A = \frac{1}{2}$ due: e^x term

$-B = 1 \Rightarrow B = -1$ due x term

$-D = 0$, $D = 0$

代入

$y_p = \frac{1}{2} x e^x - x$

Answer: $y_p = \frac{1}{2} x e^x - x$

First Order Derivative(8)

This is in the standard form y' + P(x)y = Q(x), where:

⑧(Exponential forcing)

$$y' + y = e^{2x}$$

$$P(x) = 1$$

$$Q(x) = e^{2x}$$

First Order Derivative(8)

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First Order Derivative(8)

2. Find the Integrating Factor

The integrating factor, often denoted as (below), is calculated using the formula:

$$\mu(x) = e^{\int P(x)dx}$$

Substituting P(x) = 1:

$$\mu(x) = e^{\int 1dx} = e^x$$

First Order Derivative(8)

3. Multiply and Integrate

Multiply every term in the original differential equation by e power of x:

$$e^x y' + e^x y = e^x e^{2x}$$

$$e^x y' + e^x y = e^{3x}$$

The left side is now the derivative of the product ($e^x y$) by the product rule:

$$\frac{d}{dx}(e^x y) = e^{3x}$$

First Order Derivative(8)

Now, integrate both sides with respect to x:

$$\int \frac{d}{dx}(e^x y)dx = \int e^{3x} dx$$

$$e^x y = \frac{1}{3} e^{3x} + C$$

(Where C is the constant of integration)

First Order Derivative(8)

4. Solve for y

To isolate y, divide the entire equation by e^x (or multiply by e^{-x}):, integrate both sides with respect to x:

$$y = \frac{\frac{1}{3}e^{3x} + C}{e^x}$$

$$y = \frac{1}{3}e^{2x} + Ce^{-x}$$

First Order Derivative(8)

Final Solution

The general solution to the differential equation is:

$$y = \frac{1}{3}e^{2x} + Ce^{-x}$$

First Order Derivative(8)

Differential equations of this nature describe systems where a rate of change is influenced by both the current state and an external stimulus. This specific mathematical structure is foundational across several scientific and engineering domains.

1. Thermodynamics and Cooling

In thermal physics, this equation represents the relationship between an object’s temperature and its environment. When a heat source increases in intensity over time, the equation tracks how the object’s internal energy responds to that external thermal pressure while simultaneously losing heat to the surroundings.

First Order Derivative(8)

2. Fluid Dynamics and Concentration
In chemical engineering, this model describes the concentration of a substance within a mixing tank. It captures the balance between a steady outflow of the mixture and a rapidly increasing inflow of a specific reactant, allowing for the prediction of how quickly a solution reaches a particular chemical density.
3. Electrical Circuit Analysis
Within elective systems, particularly those containing resistors and inductors, this equation governs the flow of current. The exponential component represents a "forcing function"—a power source that is ramping up—while the internal variables represent the resistance and storage of energy within the circuit components.

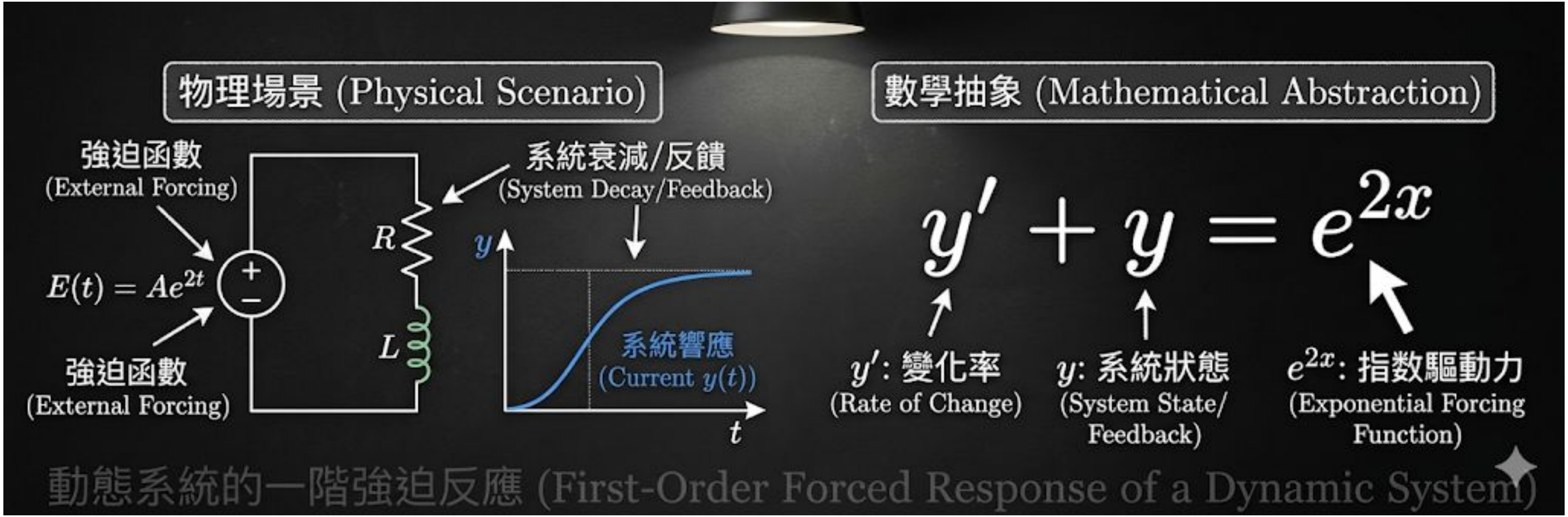
First Order Derivative(8)

1. 核心物理概念：輸入與響應
在抽象層面，這個方程式可以被理解為：
y' (變化率)：系統當前狀態改變的速度。
y (系統衰減/回饋)：系統根據當前狀態 產生的自然反應（如冷卻、摩擦或消耗）。
e^{2x} (指數驅動力)：外界施加的一個隨時間迅速增強的影響力。

First Order Derivative(8)

2. 具體應用領域
🌡️ 熱力學 (牛頓冷卻定律)
當一個物體放在溫度不斷升高的環境(如步入式加熱爐)中時，該方程式描述了物體溫度的變化。
左側項：代表物體因溫差產生的熱交換。
右側項：代表外界環境溫度呈指數級上升。
⚡ 電路分析 (RL 電路)
在包含電阻(R)和電感(L)的電路中，如果接通一個指數增長的電壓源：
y：代表電路中的電流。
y'：代表電感器產生的自感電動勢(阻礙電流變化)。
右側項：代表外部電源提供的隨時間增強的電動勢。

First Order Derivative(8)



Second Order Derivative(10)

$$y'' - y = e^x + x$$

First, solve the corresponding homogeneous equation:

$$y'' - y = 0$$

The characteristic equation is:

$$r^2 - 1 = 0 \implies (r-1)(r+1) = 0$$

The roots are .$r_1 = 1$ and $r_2 = -1$

Therefore:

$$y_h = C_1 e^x + C_2 e^{-x}$$

Second Order Derivative(10)

$$y'' - y = e^x + x$$

2. Find the Particular Solution (y_p)

We look at the non-homogeneous part:

$$g(x) = e^x + x$$

. We split this into two parts: $g_1(x) = e^x$ and $g_2(x) = x$

For : $g_1(x) = e^x$

Normally, we would guess Ae^x . However, e^x is already a solution to the homogeneous equation (it's in y_h). To avoid redundancy, we multiply by x:

$$y_{p1} = Axe^x$$

Second Order Derivative(10)

$$y'' - y = e^x + x$$

For $g_2(x) = x$

This is a first-degree polynomial. We guess:

$$y_{p2} = Bx + D$$

Combining them, our total guess is **:**

$$y_p = Axe^x + Bx + D$$

Second Order Derivative(10)

$$y'' - y = e^x + x$$

3. Solve for Constants

Now, find the derivatives of y_p and plug them back into the original ODE $(y'' - y = e^x + x)$.

$$y_p = Axe^x + Bx + D$$

$$y_p'' = A(e^x + e^x + xe^x) = A(2e^x + xe^x)$$

$$y_p' = A(e^x + xe^x) + B$$

Substitute into $y'' - y$:

$$(2Ae^x + Axe^x) - (Axe^x + Bx + D) = e^x + x$$

$$2Ae^x - Bx - D = e^x + x$$

Second Order Derivative(10)

$$e^x \textbf{ term: } 2A = 1 \implies A = \tfrac{1}{2}$$

$$x \textbf{ term: } -B = 1 \implies B = -1$$

$$\textbf{Constant term: } -D = 0 \implies D = 0$$

$$y'' - y = e^x + x$$

So, the particular solution is:

$$y_p = \frac{1}{2} x e^x - x$$

$$y'' - y = e^x + x$$

4. General Solution

Combine the homogeneous and particular solutions:

$$y = C_1 e^x + C_2 e^{-x} + \frac{1}{2} x e^x - x$$

Second Order Derivative(10)

$$y'' - y = e^x + x$$

$$y = C_1e^x + C_2e^{-x} + \frac{1}{2}xe^x - x$$

3. Practical Modeling Scenarios

Application	$y'' - y$ (System)	$e^x + x$ (Input)
Circuit Theory	The resistance and capacitance of a loop.	A power source increasing exponen over time.
Heat Transfer	The steady-state cooling property of a material.	An external heat source that is accelerating.
Control Systems	The feedback loop of a mechanical arm.	The command signal telling the arm to move.

Second Order Derivative(10)

Summary of the Method

The method of **Undetermined Coefficients** concludes that for any linear system, the total behavior is simply the sum of its "natural self" and the "influence of its environment." By identifying the overlap between the two (as we did with e^x), we can predict whether the system will remain stable or grow toward infinity.

Second Order Derivative(10)

1. 內在特徵與外部影響的總和

這個方程告訴我們，任何系統的最終狀態(總解)都是由兩個部分組成的：

- **系統的「本生反應」**：這是系統在沒有外界干擾時，根據其自身結構(例如彈簧的剛性或電路的配置)所表現出的自然傾向。
- **系統的「被迫反應」**：這是系統為了應對外界持續施加的力量(如電力脈衝或物理推力)而 產生的特定行為。**結論：** 最終結果並非兩者擇一，而是兩者共同疊加後的 產物。

2. 當「外力」與「本性」產生共振

在求解過程中，我們發現外力中包含了一個與系統本性完全相同的頻率或趨勢(例如那個特殊的指數增長項)。這在物理意義上代表了 **共振(Resonance)**。

- 當外界輸入的節奏與系統 內在的節奏同步時，系統的反應就不再只是單純的跟隨，而是會隨著時間的推移而**自我放大** 。
- 這在工程應用中是一個重要的警訊：它預示著系統可能會因為能量累積過快而變得不穩定，甚至導致結構損壞。

Second Order Derivative(10)

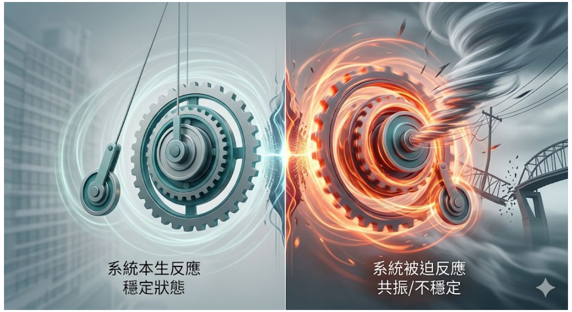
- **3. 預測系統的長期趨向**

透過這種抽象的分析方式，工程師不需要實際建造出原型，就能預知系統的未來：

- **穩定性分析**：如果系統的「本性」是趨於收斂的，但「外力」是發散的，那麼長期來看系統將會失控。
- **精準控制**：我們可以透過調整系統的參數（例如改變材料性質），來抵銷掉不想要的外力影響，使系統最終趨向於我們預期的目標。

總結

這類方程的應用結論在於：**它提供了一套邏輯框架，讓我們能精確區分哪些行為是系統「天生」的，哪些是「環境」造成的。** 這對於設計航太飛行器、精密懸吊系統或是電子通訊設備來 說，是確保其安全與精準運作的基石。



2026/3/2可分離微分方程 separable equation

$$y' = \frac{dy}{dx}$$

某量y的變化率由目前某x量控制，則以y’表示。

2026/3/2可分離微分方程 separable equation

若一微分方程如下,試求其通解及滿足初始值條件之特解:

$y' = 5x^4 + 2,\ y(0) = 3$

Answer:

$\frac{dy}{dx} = 5x^4 + 2$

上式可分離為

$dy = (5x^4 + 2)\,dx$

兩邊積分可得其通解

1/2

$y = \int (5x^4 + 2)$

$= \frac{1}{5} \times 5x^{4+1} + 2x + C$

$= x^5 + 2x + C$

若 $y(0) = 3$, 將 0 代入 x (通

$y = 0^5 + 2 \cdot 0 + C = 3$

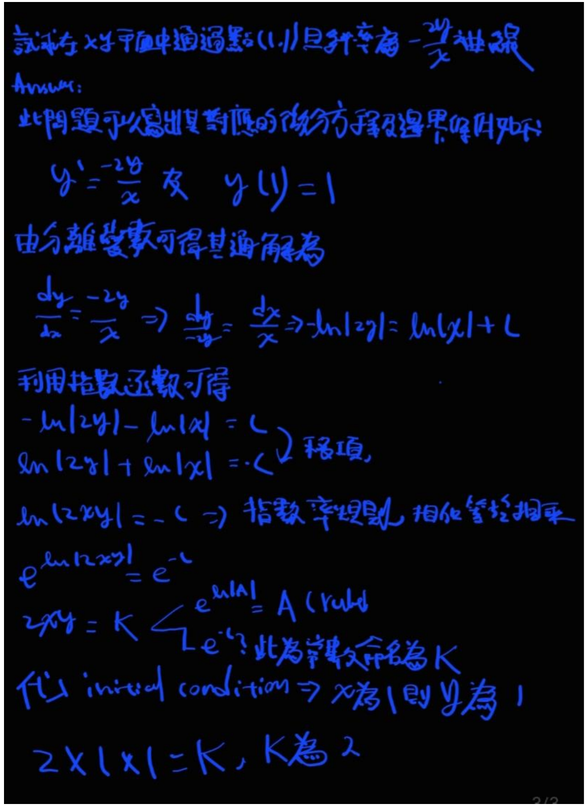
∴ 則 $C = 3$ 及其特解是

$y = x^5 + 2x + 3$

微分方程 到 積分後的方程 得通解General Solution

積分後的方程 到 帶入初始值條件initial condition 得特解particular solution

2026/3/2可分離微分方程 separable equation



試求在 xy 平面中通過點(1,1)且斜率為 $-\frac{2y}{x}$ 之曲線

Answer:

此問題可以寫出其對應的微分方程及邊界條件如下:

$y' = \frac{-2y}{x}$ 及 $y(1) = 1$

由分離變數可得其通解為

$\frac{dy}{dx} = \frac{-2y}{x} \Rightarrow \frac{dy}{-2y} = \frac{dx}{x} \Rightarrow -\ln|2y| = \ln|x| + C$

利用指數函數可得

$-\ln|2y| - \ln|x| = C$

$\ln|2y| + \ln|x| = -C$ 移項,

$\ln|2xy| = -C \Rightarrow$ 指數率規則, 相加等於相乘

$e^{\ln|2xy|} = e^{-C}$

$2xy = K$ $e^{\ln|A|} = A$ (rule)

e^{-C}? 此為常數命名為 K

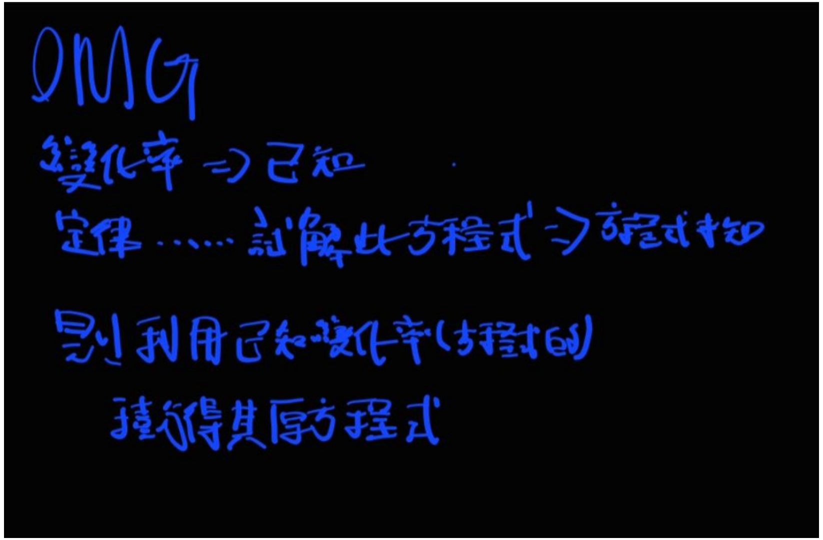
代入 initial condition $\Rightarrow$ x為1則y為1

$2 \times 1 \times 1 = K$, K為2

一曲線的斜率為某y’求其曲線y。

當變數如x,y於分母位置時可用指數率相關規則進行積分得其通解再運用指數率得其特解。

2026/3/2可分離微分方程 separable equation



用已知求未知

如 方程變化率 求 方程 用積分求解

2026/3/2可分離微分方程 separable equation

W=P.T

當W固定則Power和Time成反比

問題：實驗結果表示P的情況W瓦數的變化率為T(P)的變化率為 $-\frac{T}{P}$，此時有某方程式，試解此方程。

Answer:

$$T' = -\frac{T}{P} \Rightarrow \frac{dT}{dP} = -\frac{T}{P} \Rightarrow \frac{dT}{T} = -\frac{dP}{P}$$

$\ln|T| = -\ln|P| + C$ (通解)

$\ln|T| + \ln|P| = C$ (移項)

$\ln|TP| = C$ (指數律法則)

$e^{\ln|TP|} = e^{C} \Rightarrow TP = K$ (指數律法則)

即：Power和Time成反比
當Watt固定之時。

Wott, power and time, 當從實驗得知當W 為某固定值, 則T(P)為P的變化率公式可求T and P的變化率。。

得其方程並得以 P 和 T 成反比解釋。

End